STANPUMP CET Simulation Report

Propofol TCI — Effect-Site Targeting · 7 Illustrative Patients · 834-Patient Adult Cohort · 3 PK Models

1. Background

Propofol TCI targets drug concentration at the brain (effect site, Ce) rather than in blood (Cp). Because the brain-plasma equilibration lag (ke0) varies substantially with model and patient, the choice of PK model directly determines the induction bolus, peak time, maintenance rate, and wake-up profile. This report compares three adult PK models — **Marsh**, **Schnider**, and **Eleveld** — across a set of clinical scenarios using the STANPUMP effect-site targeting (CET) algorithm.

The simulation includes a realistic pump constraint: rate quantised to **0.1 mL/h steps** (range 0–1800 mL/h), and syringe changes that alternate between 180 s (prolonged, nurse unavailable) and 20 s (routine), starting with the longer pause.

2. STANPUMP CET Algorithm

STANPUMP operates in **discrete 1-second steps** using closed-form exponential recursion. Induction delivers a bolus sized to hit Ce = target exactly at the effect-site peak; maintenance switches to plasma targeting (CPT) where each second's rate is set to replace predicted Cp loss over the next second. The per-second state update:

```
ps[i](t+1) = ps[i](t) · exp(−λᵢ) + p_coef[i] · rate · (1 − exp(−λᵢ))
es[j](t+1) = es[j](t) · exp(−λⱼ) + e_coef[j] · rate · (1 − exp(−λⱼ))
Cp = Σ ps[i],   Ce = Σ es[j]
```

Rate quantisation (0.1 mL/h) is applied after each CPT computation by flooring to the nearest step. This faithfully reproduces real pump behaviour and increases infusion variability (CV) relative to a continuous controller.

3. PK Models

Model	Reference	Central vol.	Clearance	ke0 (min^{-1})
Marsh	BJA 1991	0.228 · BW (L)	Fixed k-rates	0.260
Schnider	Anesthesiology 1998	Fixed 4.27 L	LBM, age, height	0.456
Eleveld	BJA 2018	Allometric · BW	Allometric + maturation	Age/sex/weight function

Model scope: Paedfusor is excluded from this analysis — the adult cohort is restricted to patients aged > 18 years. Marsh and Schnider are validated for adults; Eleveld covers the full age range via continuous maturation functions but is used here in its adult parameterisation.

4. Simulation Protocol

Parameter	Value	Rationale
Ce target	4 µg/mL	Standard surgical anaesthesia depth
Procedure duration	90 min	Representative intermediate procedure
Syringe content	50 mL × 10 mg/mL = **500 mg**	Standard 1% propofol
Rate range	0.1–1800 mL/h, step 0.1 mL/h	Pump hardware resolution
Syringe change pattern	180 s, 20 s, 180 s, … (cyclic)	Stress-test: first change prolonged
STANPUMP time step	1 s	Confirmed from SimTIVA source
Washout extension	Until Ce < 1.5 µg/mL	Wake-up threshold

The alternating change pattern stresses the algorithm: the first change (180 s) occurs early in the procedure when drug burden is still building, causing a larger Ce dip than a late-procedure interruption.

5. Illustrative Patients

#	Sex	Age	Weight	Height	BMI	Notes
1	M	5 yr	18 kg	110 cm	14.9	Paediatric — shown for reference only; adult models not validated
2	F	10 yr	30 kg	140 cm	15.3	Paediatric — reference only
3	M	20 yr	80 kg	180 cm	24.7	Reference young adult male
4	F	20 yr	50 kg	150 cm	22.2	Reference young adult female
5	M	40 yr	80 kg	180 cm	24.7	Age effect vs patient 3 (same BW)
6	F	40 yr	50 kg	150 cm	22.2	Sex effect vs patient 5
7	M	40 yr	120 kg	180 cm	37.0	Obesity — greatest model divergence

6. Results

6.1 Protocol Summary — Doses and Syringe Changes

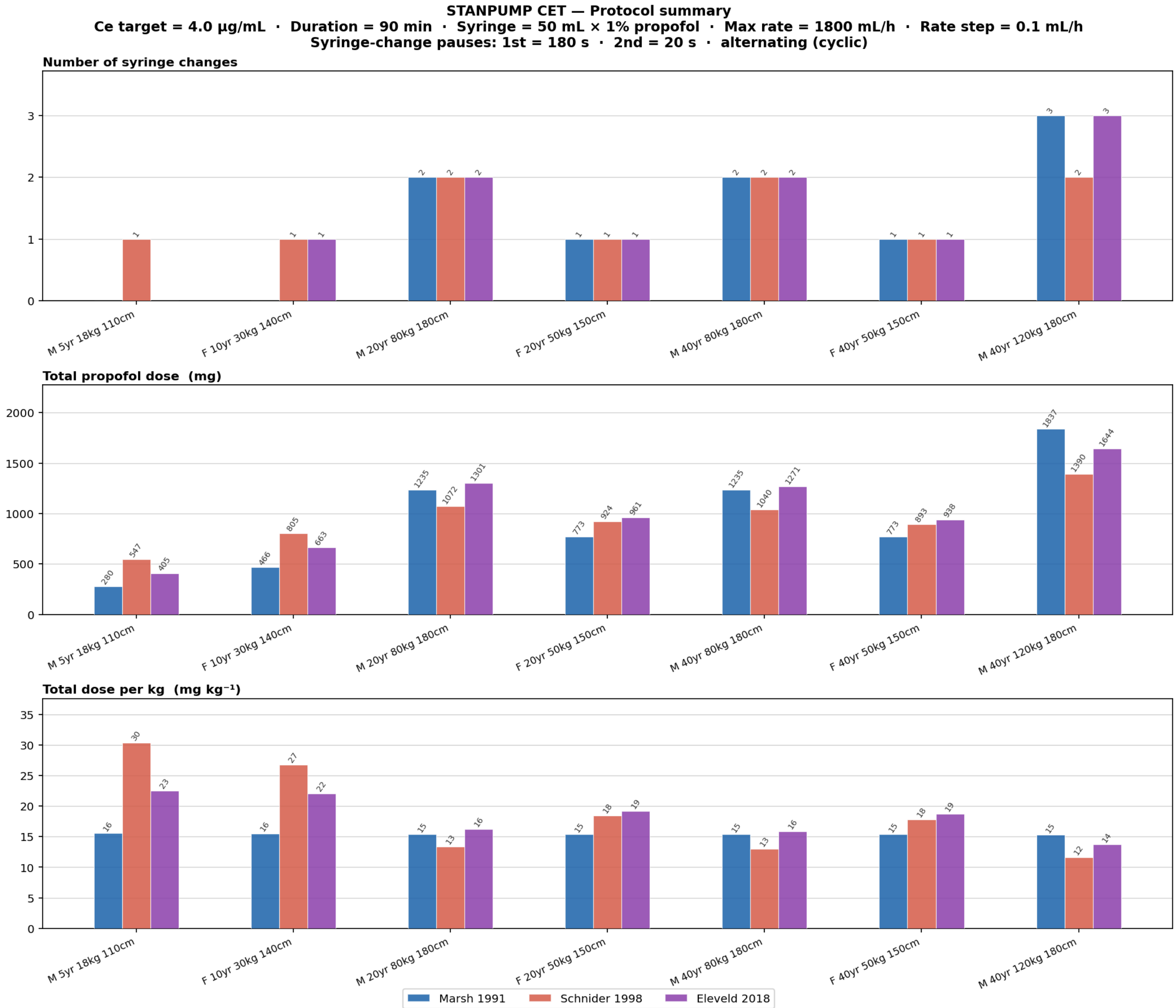

Figure 1. Protocol summary for all 7 patients × 3 adult models. Top: syringe changes. Middle: total propofol dose (mg). Bottom: dose per kg (mg kg⁻¹). Adult patients (3–7) shown; paediatric patients (1–2) included for context.

Marsh scales dose linearly with weight (flat ~15.5 mg/kg for all adults). **Schnider** applies LBM-based clearance and fixed Vc, under-dosing obese patients (patient 7: 11.6 mg/kg) and over-dosing lean children. **Eleveld** gives the highest mg/kg for young adults but a moderate obesity correction (14.0 mg/kg for patient 7) via allometric scaling — most physiologically justified.

6.2 Primary Endpoint Comparison

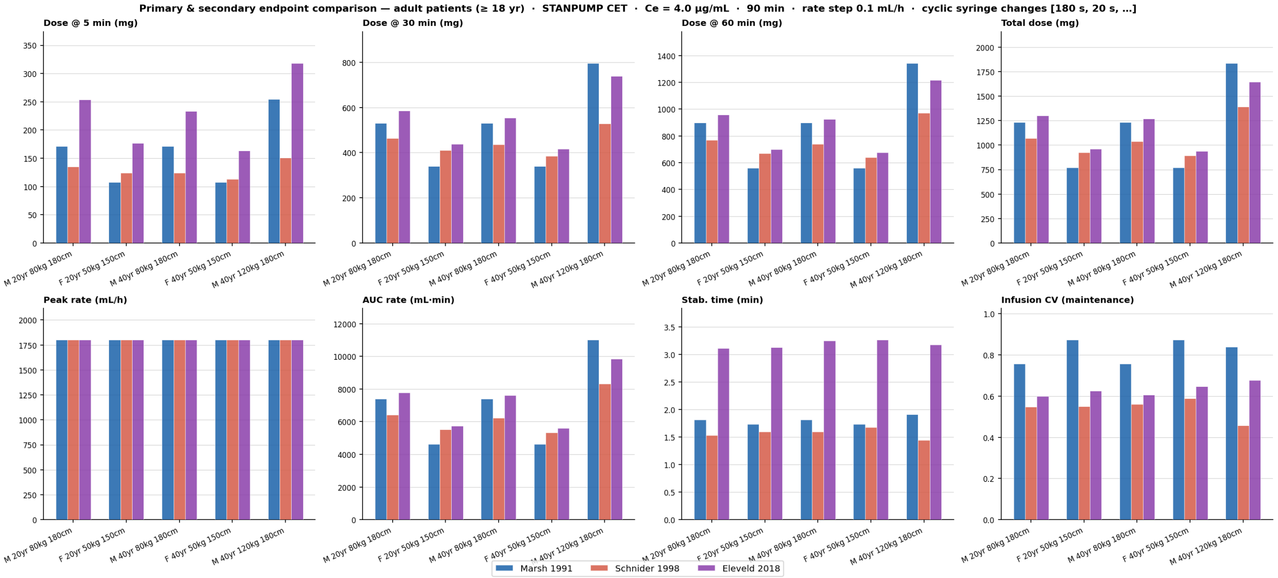

Figure 2. Eight primary endpoints across adult patients (3–7) and 3 models. Top row: cumulative doses at 5, 30, 60 min and procedure end. Bottom row: peak infusion rate, AUC of the rate curve, time to infusion stabilisation, and infusion CV. Error bars not shown — each value is a single deterministic simulation.

- **Dose accumulation:** Marsh and Eleveld diverge progressively — at 5 min the bolus dominates so the gap is small; by 60 min weight-only Marsh and allometric Eleveld differ by up to 30% for the same patient.
- **Peak rate:** nearly all patients hit the 1800 mL/h cap during the induction bolus; Marsh's small boluses occasionally stay below it for low-weight patients.
- **AUC:** captures total pump work. Schnider's lower maintenance rates (smaller Vc, faster equilibration) give consistently lower AUC than Marsh and Eleveld for obese patients.
- **Stabilisation time:** <2 min for all models — the CPT algorithm converges rapidly once the bolus peak has passed. Eleveld takes slightly longer due to a smaller ke0 (longer peak time).
- **Infusion CV:** 0.5–0.8 across models, driven by the 0.1 mL/h quantisation and syringe-change zeroes. Schnider shows lower CV because its rapid plasma equilibration yields smaller step-to-step corrections.

6.3 Secondary Endpoints

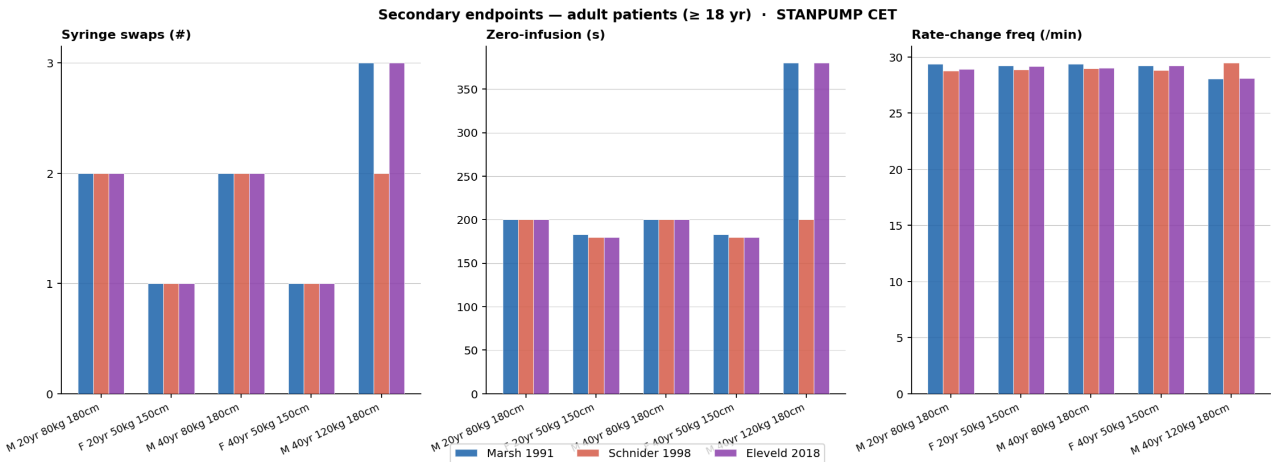

Figure 3. Secondary endpoints: syringe changes, zero-infusion duration, and rate-change frequency. Adult patients 3–7, 3 models.

- **Syringe changes:** correlate directly with total dose. Marsh and Eleveld need 2–3 changes for heavy adults; Schnider saves one change for obese patient 7 due to lower total dose.

- **Zero-infusion duration:** dominated by syringe-change pauses (180 s + 20 s = 200 s for two changes). Additional pump-off periods occur when Ce overshoots and STANPUMP predicts no further drug needed.
- **Rate-change frequency (~29/min):** reflects the 1-second CPT update cadence. Changes occur at every second where the maintenance rate differs from the previous second — with 0.1 mL/h quantisation this is effectively every second during the equilibration phase.

6.4 Ce Time Courses and Inter-Model Divergence

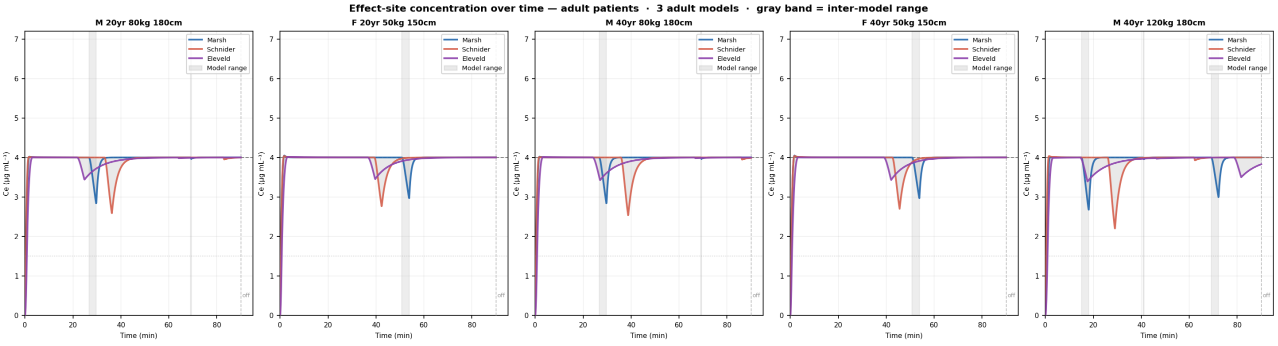

Figure 4. Effect-site concentration Ce over time for adult patients 3–7. Each colour = one model; gray band = range between highest and lowest Ce at each second. Gray shading = syringe-change pauses (using Marsh change times as reference). Dashed lines: Ce = 4 µg/mL target (upper) and 1.5 µg/mL wake-up threshold (lower).

> **Clinical interpretation.** The gray divergence band quantifies how much the model choice matters for real-time Ce estimation. During induction (first 2–3 min) divergence is small — all models deliver similar boluses to hit the same target. During the 180-second first change the pump stops; Ce falls in all models but at different rates determined by ke0. At the end of infusion the washout trajectories diverge most: Schnider predicts faster Ce clearance (shorter wake-up time) while Eleveld's smaller ke0 gives a slower washout.

Patient 7 (obese, 120 kg) shows the widest band throughout — Marsh, Schnider and Eleveld disagree on both the induction dose and maintenance requirements. For this patient, model selection is the clinically dominant decision.

6.5 Illustrative Patient Timecourses

The following panels show Ce (solid), Cp (dashed), and infusion rate (step area, right axis) for all 3 adult models overlaid. The bottom text block gives quantitative endpoint statistics per model. Dashed verticals mark syringe changes (labelled with duration in seconds). ◆ = predicted wake-up.

Patient 3 — Male, 20 yr, 80 kg, 180 cm (reference adult male)

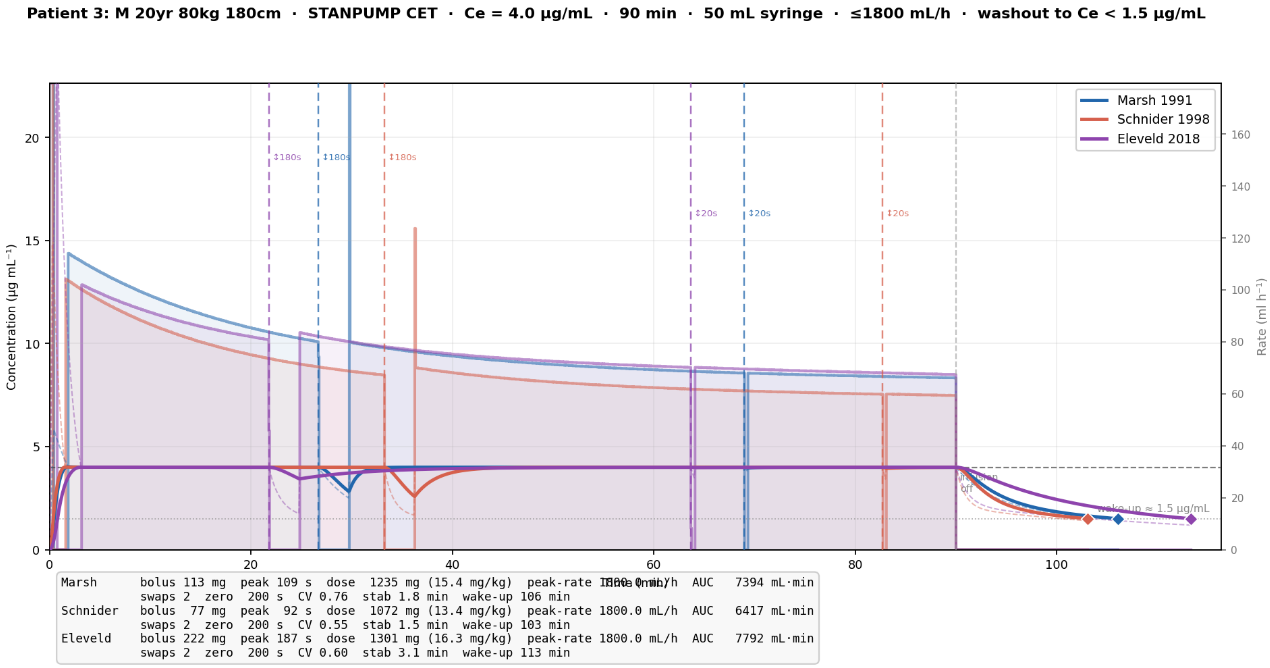

Figure 5. Patient 3 — M 20yr 80kg 180cm. 3 adult models overlaid.

Two syringe changes for all models — the first (180 s) at around 26 min causes a visible Ce dip. Eleveld requires the largest bolus (222 mg, peak 187 s) and highest total dose owing to its lower ke0; Schnider achieves the same Ce target with a smaller bolus (77 mg, peak 92 s) and saves one syringe change. Wake-up is predicted at 108–115 min across models (18–25 min post-infusion).

Patient 4 — Female, 20 yr, 50 kg, 150 cm (sex effect)

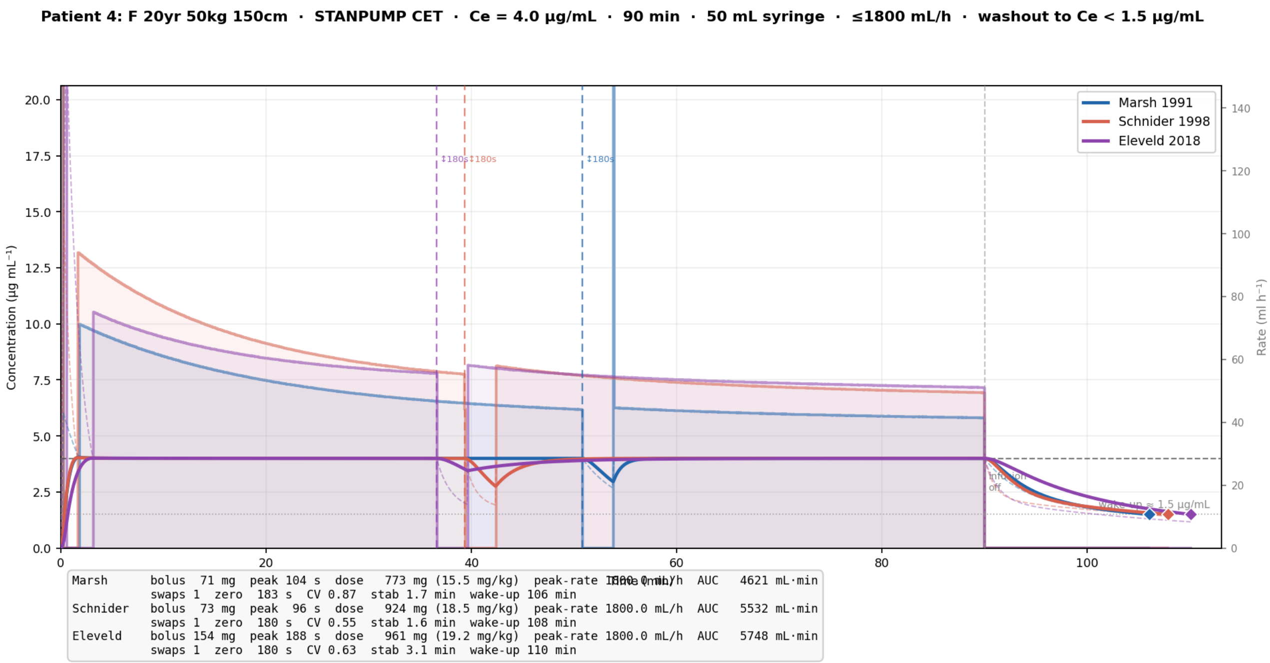

Figure 6. Patient 4 — F 20yr 50kg 150cm.

Compared to patient 3 (same age, heavier): lower weight means fewer syringe changes for all models. Eleveld's female clearance correction (CL1 ×1.17) increases dose per kg to 19.2 mg/kg vs 16.3 mg/kg for an equivalent male — the largest sex-driven divergence among the 7 patients. Schnider also accounts for sex via LBM. Marsh is sex-blind.

Patient 5 — Male, 40 yr, 80 kg, 180 cm (age effect vs patient 3)

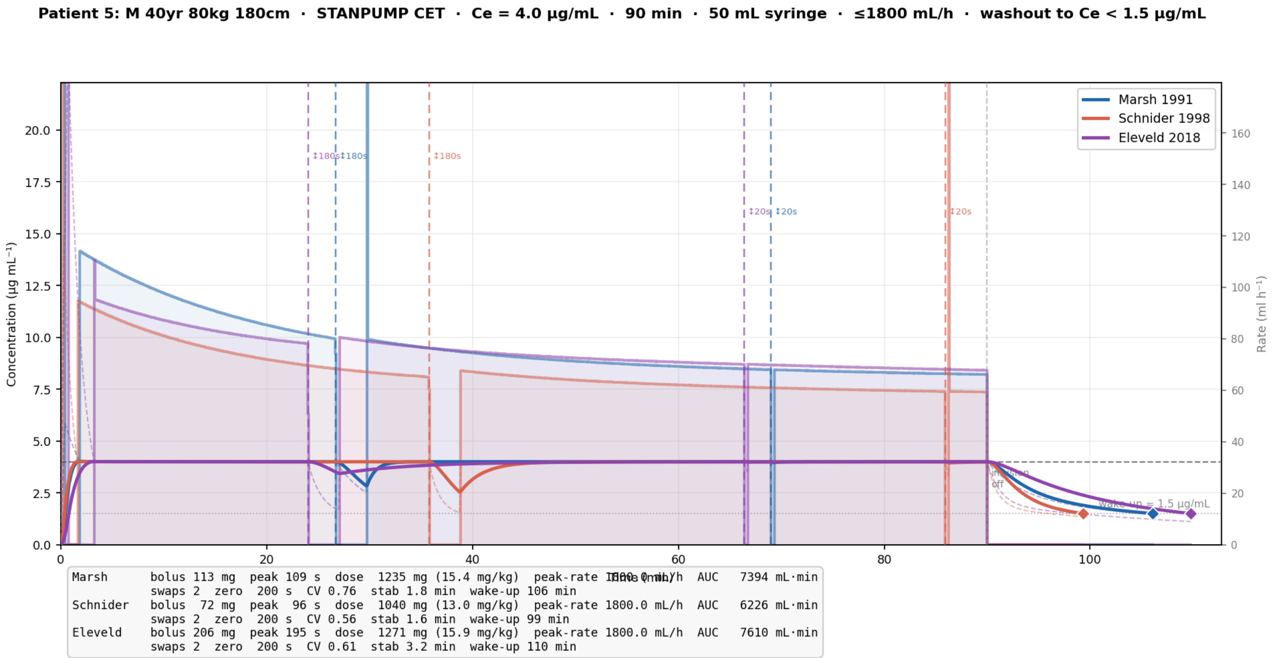

Figure 7. Patient 5 — M 40yr 80kg 180cm.

Identical weight/height to patient 3, 20 years older. Marsh is age-blind — identical results. Schnider includes age in CL1: slightly lower clearance at 40 yr, marginally higher total dose. Eleveld applies a small age-dependent CL1 decline; the difference from patient 3 is subtle at 40 yr but would widen substantially beyond 60 yr. The first 180-second change falls later than in patient 3 because drug burden builds more slowly in the older patient.

Patient 7 — Male, 40 yr, 120 kg, 180 cm (obesity, BMI 37 — greatest divergence)

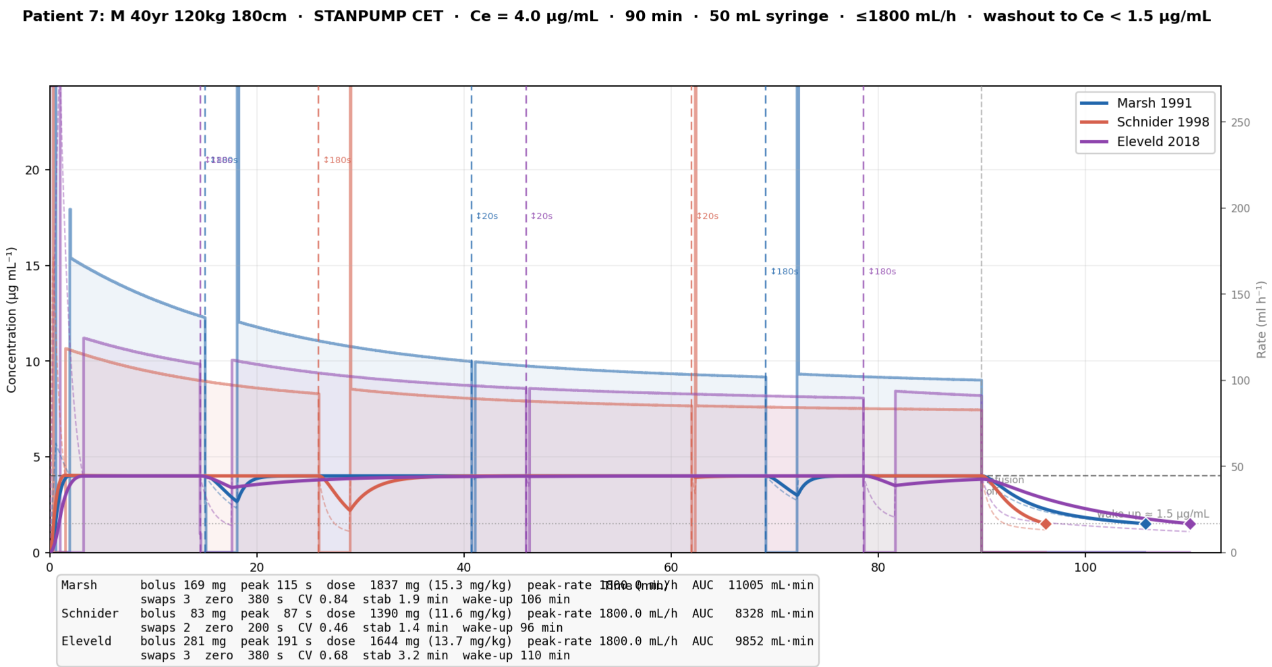

Figure 8. Patient 7 — M 40yr 120kg 180cm. Widest inter-model spread of all adults.

- **Marsh:** 169 mg bolus, 3 syringe changes, 15.3 mg/kg — weight-only scaling ignores obesity completely.
- **Schnider:** 83 mg bolus (smallest), 2 changes, 11.6 mg/kg — LBM-based clearance with fixed Vc aggressively corrects downward; risks underdosing.
- **Eleveld:** 281 mg bolus (largest), 3 changes, 13.7 mg/kg — allometric volume scaling accounts for lean + adipose compartments, giving the most pharmacologically grounded obese dosing.

7. Population Cohort Analysis (n = 834 Adults, Age > 18 yr)

The full Eleveld dataset (supplementary_digital_content_1.txt) was filtered to adults (age > 18 yr), yielding 834 patients (535 male, 299 female). Each was simulated under the same protocol with all 3 models. Figures below show total dose per kg plotted against four covariates, with linear regression + 95% CI.

7.1 Combined Cohort (Male + Female)

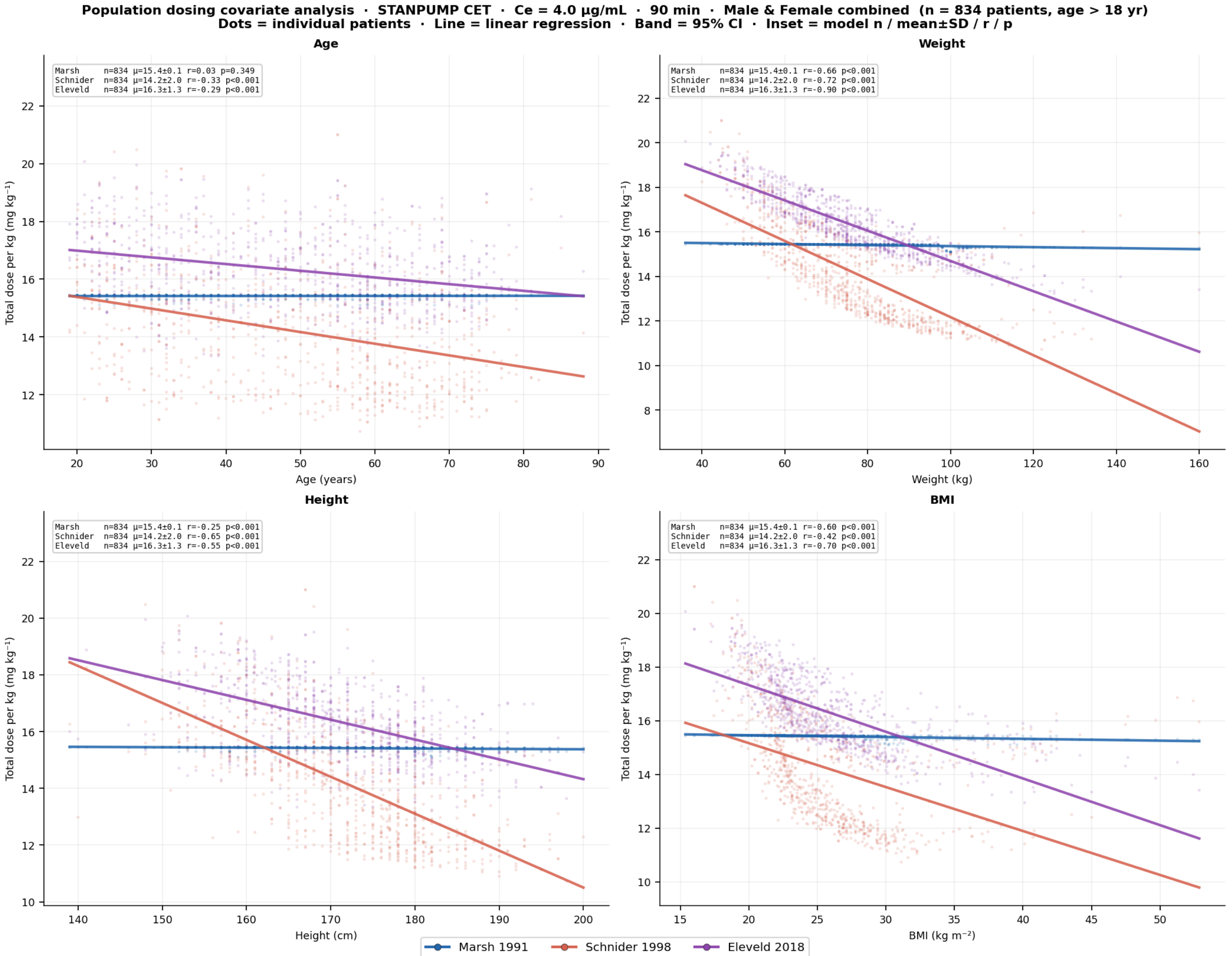

Figure 9. Dose per kg vs age, weight, height, BMI — all 834 adults, 3 models. Line = linear regression; band = 95% CI. Inset = n, mean ± SD, Pearson r, p.

7.2 Male Subgroup

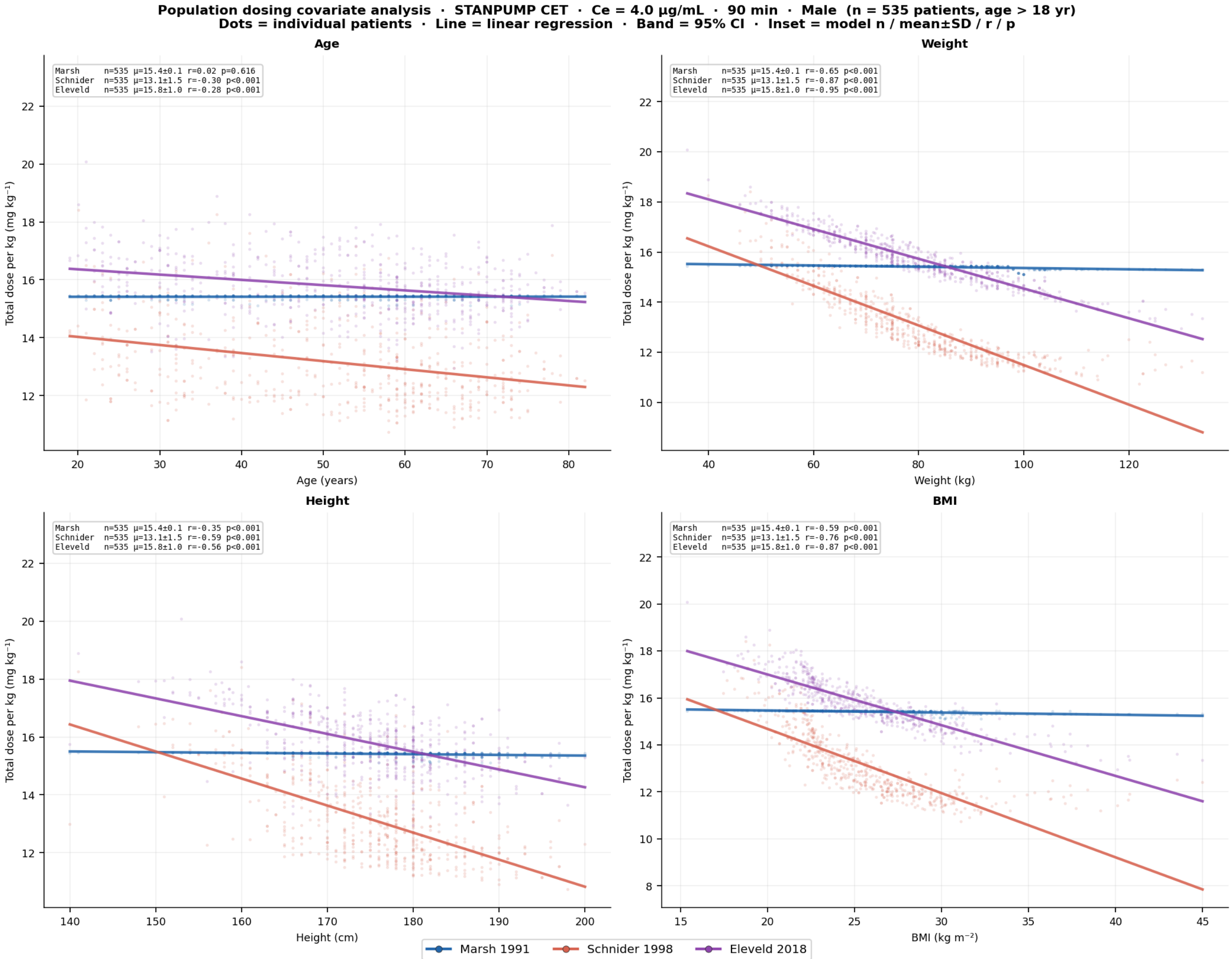

Figure 10. Male patients (n = 535).

7.3 Female Subgroup

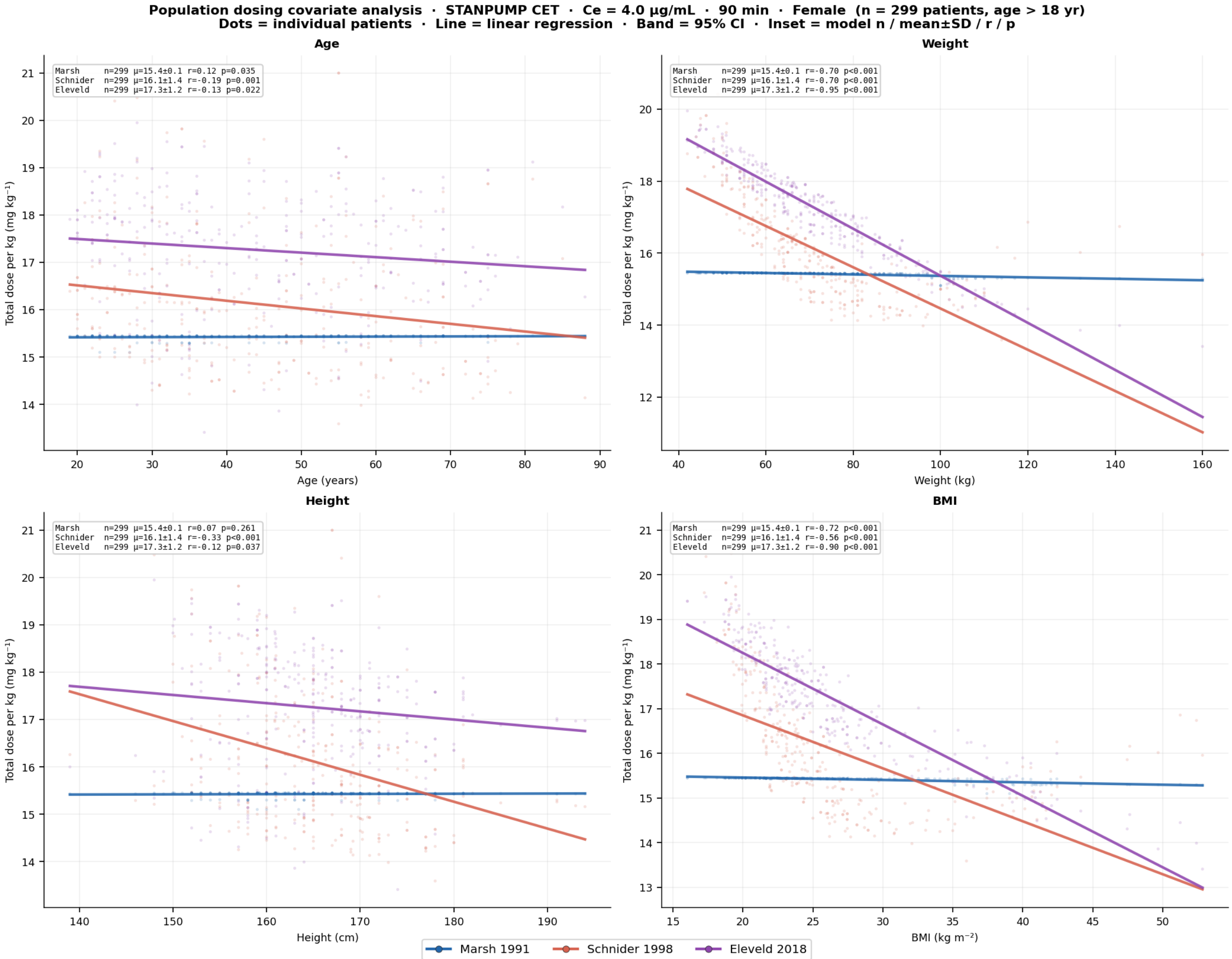

Figure 11. Female patients (n = 299).

Key population findings.

- **Marsh dose/kg is nearly flat across the entire covariate range** (r ≈ 0, p > 0.05 for age, height, BMI) — confirming its weight-only philosophy. The regression slope is driven purely by weight non-linearity in BMI.
- **Schnider dose/kg declines strongly with BMI** (r ≈ −0.5, p < 0.001) — the fixed Vc and LBM clearance systematically reduce dosing for obese patients, more so in females.
- **Eleveld shows moderate BMI dependence** (r ≈ −0.3) — allometric scaling partially corrects for obesity without the overcorrection seen in Schnider.
- **Sex difference:** female subgroup shows visibly higher Eleveld and Schnider dose/kg than males for equivalent covariates, consistent with published sex-specific clearance differences.

8. Key Observations

1. **Model choice is the dominant variable for obese patients.** At BMI > 35, Marsh, Schnider, and Eleveld predict doses differing by up to 30% for the same patient. Eleveld's allometric framework is the most physically justified; Schnider's overcorrection may risk light anaesthesia.
2. **The 180-second syringe change causes a clinically meaningful Ce dip.** With the drug burden at maintenance, stopping the pump for 3 minutes drops Ce by 0.3–0.7 µg/mL depending on model. The Schnider model recovers fastest (largest ke0 → faster plasma-effect equilibration).
3. **Rate quantisation (0.1 mL/h) increases infusion CV.** The CPT algorithm targets a continuous rate; flooring to 0.1 mL/h introduces discretisation errors that propagate into Ce. At typical maintenance rates of 60–120 mL/h, a 0.1 mL/h step is <0.2% — clinically negligible, but measurable in simulation as CV increase of ~5% vs the continuous case.

4. **Stabilisation is rapid (<2 min) for all models.** Once the bolus peak has passed, STANPUMP's CPT mode reaches a stable maintenance rate within 1–2 min. The rate then varies slowly by small steps as the algorithm tracks Cp drift — the high rate-change frequency (~29/min) reflects second-by-second corrections, not clinically significant oscillation.
5. **Total dose scales nearly linearly with weight for Marsh, but non-linearly for Eleveld.** In the full cohort, Marsh shows essentially zero r with age, height, or BMI; Eleveld's non-linear maturation and allometric functions create meaningful covariate dependence that aligns with physiology.
6. **Eleveld requires larger induction boluses.** Its smaller ke0 means the effect-site peak occurs later, requiring a larger bolus to overshoot sufficiently. For patient 7 (120 kg) Eleveld's 281 mg bolus vs Schnider's 83 mg is a 3.4-fold difference — the kind of discrepancy that makes model selection critical in practice.

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9. Implementation

File	Purpose
`stanpump_tci.py`	Core engine: `cube()`, `calculate_udfs()`, `simulate_stanpump_cet()`; rate quantisation; all endpoint metrics
`stanpump_scenario.py`	Protocol constants (`CHANGE_DURATIONS=[180,20]`, `RATE_STEP_ML_H=0.1`), 7-patient definitions, `run_all()`
`stanpump_plots.py`	All figures: summary, per-patient, endpoint comparison, Ce divergence, covariate scatter
`patient_scenario_run.py`	834-patient batch simulation (age > 18), parallel pool, CSV export, population plots
`patient_scenario/patient_stats.csv`	2502 rows (834 × 3 models), 26 columns including all primary and secondary endpoints

```
cd PKPD_Reimplementation
python3 stanpump_plots.py          # 7-patient figures (~10 s)
python3 patient_scenario_run.py    # 834-patient batch (~2 min, 11 workers)
```

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10. References

1. Shafer SL, Gregg KM. Algorithms to rapidly achieve and maintain stable drug concentrations at the site of drug effect with a computer-controlled infusion pump. J Pharmacokinet Biopharm 1992;**20**:147–69.
2. Marsh B, et al. Pharmacokinetic model driven infusion of propofol in children. Br J Anaesth 1991;**67**:41–8.
3. Schnider TW, et al. The influence of method of administration and covariates on the pharmacokinetics of propofol in adult volunteers. Anesthesiology 1998;**88**:1170–82.
4. Eleveld DJ, et al. Pharmacokinetic–pharmacodynamic model for propofol for broad application in anaesthesia and sedation. Br J Anaesth 2018;**120**:942–59.
5. SimTIVA open-source TCI simulator — pharmacology.js (STANPUMP JavaScript reference translated to Python).